

RESEARCH ARTICLE

CERVIA: A Hybrid Deep Learning Framework With Explainable AI for Automated Cervical Cancer Classification From Pap Smear Images

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ABSTRACT Cervical cancer is a primary health concern worldwide. The World Health Organization (WHO) estimated that there were almost 350,000 deaths caused by the disease in 2020. Early detection by means of a Pap smear is vital. However, manual examination is prone to inter observer variability and is very time consuming. Deep learning techniques provide a potential solution. Hence, this research presents CERVIA, a hybrid deep learning network combining InceptionResNetV2 and MobileNetV3 architectures by feature fusion strategy to achieve higher discriminative power across NILM, LSIL, and HSIL categories. The problem of unbalanced classes is solved by Conditional Generative Adversarial Networks, so that the dataset becomes balanced for all the classes. Two explainable AI methods Local Interpretable Model agnostic Explanations (LIME) and SHapley Additive exPlanations (SHAP) are used to give clear interpretability, with SHAP being more closely aligned with the diagnostic features. A clinical report generation component converts the predictions into the real world information. CERVIA has been able to get 97.1% accuracy, 96.7% precision, 97.1% recall, 96.9% F1 score, and 0.99 ROC AUC on the CRIC test set thus, performing far better than the most advanced methods. Independent validation with the Brown Multicellular ThinPrep Database was also able to confirm the resulting high accuracy of 96.8%, hence, generalization is very robust. The amalgamated framework sets up a comprehensive solution that a medical professional can employ in various clinics and hospitals, and other healthcare settings.

INDEX TERMS Cervical cancer classification, deep learning, hybrid CNN architecture, explainable AI, CGAN data augmentation, pap smear analysis, medical image classification, clinical decision support.

I. INTRODUCTION

Cervical cancer is a public health problem that disproportionately affects low- and middle-income nations. Newly diagnosed female cervical cancers in 2022 amounted to roughly 662,000 across the world, which is fourth in prevalence [1]. Fatalities due to this disease are incredibly high, with 350,000 deaths reported in 2022, where 94% were reported to have occurred in low- and middle-income nations [2]. Incidence across the globe is predicted to be 760,082 new cases and 411,035 deaths by 2030 at present rates, a 14.8% and 17.8% increase respectively [3]. Early detection of precancer and malignant cervical lesions is integral to successful intervention due to the fact that

early detection significantly improves patient outcome with a significant reduction in mortality rates. Nevertheless, conventional cytopathological examination techniques are plagued with issues related to inter-observer variability, labor-intensive manual screening, and a lack of qualified pathologists in resource-limited settings.

The use of deep learning has really changed how cervical cancer is screened. Earlier, most screenings were done manually under a microscope by cytopathologists, and the results often varied. Sometimes the sensitivity was as low as 50%, sometimes up to 80%, depending on the person doing it. With deep learning, however, the system can look closely at the cell structures and pick up unusual patterns far more consistently and accurately. Automatic deep learning-based systems yield objective, reproducible findings that evaluate hundreds or even thousands of specimens per hour with

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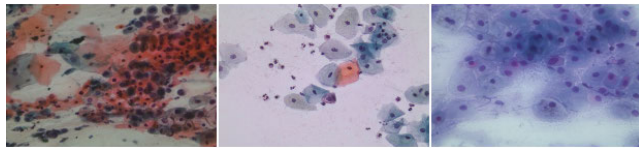


FIGURE 1. Sample images from the dataset.

excellent diagnostic ability. They learn deep hierarchical feature representations directly from unprocessed image data and identify subtle morphological variations that are missed even by skilled human observers. Deep learning models can also be situated in underprivileged areas where there is little availability of subspecialist pathologists.

This work takes advantage of the CRIC Cervix Collection, a richly annotated dataset purposely assembled for studies on cervical cancer and computer vision/machine learning. The collection emphasizes 9842 high-definition Pap smear scans at 1376×1020 pixels with 11,534 manually annotated cervical cells classified along the Bethesda System. Pap smear or Papanicolaou testing is a screening procedure where a brush or spatula is used to scrape off cells from the outer surface of the cervix that are then spread on a glass slide and coloured prior to microscopic analysis. The CRIC collection covers the complete range of cytological findings for cervical cytology, ranging from normal epithelial cells to precancerous lesions at various grades up to invasive carcinoma. Each cell was manually labeled with great care by skilled cytopathologists to provide high-quality annotation for supervised learning approaches. Some images from the dataset have been shown in Fig. 1.

This paper presents Cervical Examination and Recognition Intelligent AI abbreviated as CERVIA, a new hybrid deep learning network that combines two different architectures of convolutional neural network in a synergistic way and achieves an accuracy of 97.1% for cervical cell classification. The network classifies liquid-based cytology images into three diagnostic groups: Negative for Intraepithelial Lesion or Malignancy (NILM), Low-Grade Squamous Intraepithelial Lesion (LSIL), and High-Grade Squamous Intraepithelial Lesion (HSIL). NILM is defined as healthy, normal cervical cells, LSIL suggests mild cellular alterations that are typically due to infections with low-risk human papilloma viruses and that will often disappear spontaneously, HSIL is defined as extensive precancerous changes that must be treated promptly to avoid developing into invasive cancer.

Key contributions of this study are applying Generative Adversarial Network (GAN) based data augmentation to overcome class imbalance in the dataset as shown by Fig. 2 and achieving robust training of the model across the full spectrum of diagnostic categories. CERVIA is optimized and developed via extensive hyperparameter tuning and advanced data preprocessing techniques, establishing its superiority compared to current state-of-the-art architectures via extensive comparative analysis. Explainable artificial intelligence (XAI) methods, namely Local Interpretable Model-agnostic Explanations (LIME) and SHapley Additive exPlanations

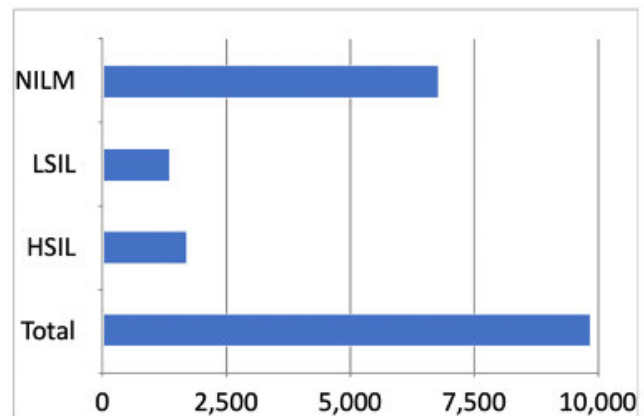


FIGURE 2. Distribution of images in different classes before GAN Augmentation.

(SHAP), are incorporated to achieve interpretable understandings of the model's decision-making processes. LIME produces localized explanations by indicating cellular areas most responsible for guiding the classification decisions, while feature importance values are computed via SHAP that quantify each pixel contribution to the ultimate prediction. XAI methods find no substitute in clinical adoption, allowing pathologists to confirm model reasoning, discover potential errors, and trust computer-aided diagnosis systems. Clinical maturity is shown through production of detailed diagnostic reports summarizing findings in clinically interpretable form.

The envisioned system is a full, transparent, and clinically deployable solution for computer-aided cervical cancer screening. By coupling high-performance classification with interpretable explanation and standardized reporting, CERVIA fills the gap between AI research and clinical implementation. The Explainability that XAI methods provide resolves a key challenge to AI adoption in clinical work, allowing doctors to verify and understand computer-assisted diagnoses prior to reaching a treatment choice.

The rest of this paper is also outlined similarly as follows. Section II reviews related works about deep learning for the diagnosis of cervical cancer and XAI applications, Section III outlines the methodology with data preprocessing, architecture of CERVIA, and extension to XAI, Section IV displays experimental results and comparison for performance, and Section V concludes with summary and future work outlines.

II. LITERATURE REVIEW

Automated cervical cancer screening has been transformed with the power of deep learning to analyze Pap smear images, overcoming the drawbacks of manual cytopathological examination such as inter-observer variability and tedious analysis. Recent research has centered on overcoming important challenges such as class imbalance, limited diversity in the dataset, and the requirement for explainable prediction in clinical practice. Developments in deep learning methods for cervical cytology have evolved since the early days of straightforward transfer learning applications to advanced

hybrid architecture that includes data augmentation, ensemble schemes, and interpretability schemes.

Overcoming severe cervical class imbalance, particularly underrepresentation of high-grade lesions, has been a primary focus of new research. Wubineh et al. applied RES_DCGAN augmentation combined with self-attention-augmented ResNet50V2 to achieve 98% binary accuracy on the Herlev and SIPaKMeD datasets, though their solution was validated on only these two testbeds and no measures of explainability were given [4]. Shandilya et al. designed a skip-connected architecture of CNNs in conjunction with GAN-based oversampling to achieve high accuracy on the five-class task on SIPaKMeD and demonstrated that skips improve feature propagation and learning efficiency, though evaluation was based on cell-type identification and excluded the clinically important NILM/LSIL/HSIL classes and measures of explainability [5]. Yang et al. constructed a high-end pipeline of Seg-UNet segmentation, multi-modal GAN-based data augmentation, and Dense CapsNet classification and achieved high accuracy on SIPaKMeD, though system computational complexity, restriction to only three coarse classes (benign/abnormal/normal), and lack of explainability measures limited its clinical utility [6]. GAN-based data augmentation offsets data sparsity by generating artificial samples of minority classes to ensure more balanced representation across diagnostic classes; however, such methods may fail to capture the morphologic variability present in populations with heterogeneous disease presentations.

Ensemble learning and hybrid architecture have appeared as effective methods to enhance classification robustness via model diversity. Gangrade et al. put forth a deep ensemble of CNN, AlexNet, and SqueezeNet, reaching 94% accuracy on five-class SIPaKMeD classification with 93.7% on seven-class Herlev data, establishing that ensemble approaches surpass lone architecture performance, although modest dataset size and lack of data augmentation constrained generalizability [7]. Siddiqui et al. presented CerviX-Ensemble, a stacking ensemble of heterogeneous CNNs with incorporation of SHAP explanations and web-app based interface, establishing state-of-the-art performance on Herlev and SIPaKMeD datasets with model interpretability in the form of explainable AI, while suffering by considerable computational overhead and lack of segmentation module incorporation [8]. Attallah created CerCan-Net, a multi-layer feature ensemble of light-weighted CNNs with transfer learning that extracted and combined features between several network depths to obtain supplementary morphological patterns, reaching good performance on SIPaKMeD, although the methodology had no explainability analysis and segmentation capabilities [9]. Sharma et al. combined DenseNet201 and InceptionV3 via fine-tuning and feature fusion, exploiting DenseNet201's effective feature reuse and InceptionV3's multi-scale hierarchical feature extraction to attain 96.54% accuracy with balanced precision (95.91%) and recall (96.44%), outperforming individual models such as ResNet-50 and Xception, although methodology failed to

utilize GAN-based data augmentation and to treat the dedicated NILM/LSIL/HSIL classification framework [10]. Although these ensemble approaches establish enhanced accuracy and robustness via model complementarity of predictions, they incur elevated computational loads and training complexity that might deter deployment in compute-limited clinical settings.

Transfer learning has been explored in a systematic manner to determine the best pretrained architecture for cervical cytology without needing to train from scratch on large scales. Kaur et al. have performed an exhaustive comparison of sixteen pretrained models such as ResNet, VGG, DenseNet, and other architectures on Herlev and SIPaKMeD datasets, where VGG16 was able to achieve high accuracy with good benchmark comparison across popular families of CNNs, although the research had no new architectural innovations, no targeted data augmentation for rare types of lesions, nor explainability mechanisms [11]. Zhang et al. have augmented DenseNet121 with Feature Pyramid Networks to enhance multi-scale feature estimation and achieved around 96% accuracy on SIPaKMeD in binary normal/abnormal versus abnormal/healthy class distinction, illustrating that architectural changes to pretrained networks enhance performance but was constrained to binary class distinction with no data augmentation nor explainability mechanisms [12]. Austin and Parvathi illustrated that cell cluster direct classification without segmentation could attain high accuracy on SIPaKMeD and challenged the traditional pipeline that mandates segmentation as a prerequisite step, although limited to binary abnormal/healthy distinction with no explainability discourse except on a very limited level [13]. Such transfer learning experiments establish seminal performance baseline measures and illustrate the potential in exploiting ImageNet-pretrained weights, albeit at the cost of usually preferring raw accuracy measures over the interpretability, fine-grained classes that matter in clinical adoption, and class imbalance robustness.

Custom-designed architectures overcoming specific technical issues within cervical image analysis have also emerged. Ji et al. used a deep U-Net ensemble to enhance segmentation performance overcoming overlapping cells, a typical issue in Pap smear microscopy, and attained cytoplasm Dice coefficients of 0.953 in comparison to 0.948 by single models, although the solution was limited to solving only segmentation and not extending to subsequent classification into diagnostic classes [14]. Zhang created GC-UNet with geometric constraints to maintain morphologically reasonable nucleus boundaries to overcome issues with overlapping and blurred cell boundaries in Pap smears, although again only solving segmentation and extending to end-to-end classification in subsequent classes [15]. Shi et al. proposed graph convolutional networks to learn spatial relationships between cellular regions while combining morphological shape features with CNN embeddings, providing an alternative representation that synthesizes both local-appearance and global-structural patterns on the UCF CC 50 dataset, although

the solution was tested in only a complete segmentation but not in a complete segmentation-classification pipeline and also on only a single data benchmark [16]. The versatility of such graph-based deep learning frameworks in handling complex morphological data extends beyond cytology, having demonstrated efficacy in broader medical interaction and rehabilitation tasks, such as described in 3D hand surface segmentation by Xing et. al. [17] and non-contact gesture recognition [29]. Mehedi et al. constructed CCanNet, a light model with only 1.3 million parameters that achieves high accuracy in SIPaKMeD while including explainability modules to confirm predictions and highlight discriminative regions, which illustrates that small models that perform competitively and are suitable even in constrained devices exist, although the solution was tested on only a single dataset with no GAN data augmentation [17]. These custom approaches make technical breakthroughs in individual aspects of the classification workflow such as segmentation performance, architectural complexity, and new feature representations, although frequently do not end up with complete, clinically translatable solutions that combine all components.

Other strategies have investigated feature fusion and high-end architectures. Zhang et al. presented DeepPap, fine-tuning ConvNets on Pap smear patch level to abolish handcrafted features, although only binary classification was addressed with no GAN augmentation or explainability [18]. Rahaman et al. created DeepCervix, combining deep features across several CNNs to attain high accuracy on class-imbalanced data, although no GAN balancing and no explainability were present [19]. Sabeena and Gopakumar combined PSPNet segmentation with ensemble classifier for four-class classification, connecting segmentation and class tasks but with no GAN fusion or XAI overlay [20]. More recent transformer-based designs have come forth, such as Yi et al.'s proposal of a multi-scale window transformer tested on vast multiclass datasets, although with no GAN balancing or interpretability overlay [21], and Sha et al.'s presentation of CerviFusionNet, a combined CNN-Transformer-GRU architecture that unifies several colposcopy modalities, although this cannot apply to cytology/LBC images and no GAN or XAI fusion exists [22]. Anand et al. revealed in skin disorder classification that architecture hybrids with GAN data imbalance and double SHAP-LIME explainability could attain 97.8% accuracy and still have interpretability [23].

In spite of great advances, severe gaps remain. Most works assess performance on few benchmarks with classifications incompatible with clinical NILM/LSIL/HSIL categories. High-performance models typically utilize computationally heavy structures with no consideration of deployment constraints and no provision of explainability mechanisms that are crucial to clinical trust. Few works incorporate successful augmentation for class imbalance, correct multi-class classification compatible with clinical practice, explainable AI to afford transparency, and creation of structured diagnostic reports. However, such complete combinations are rarely

achieved, showing that more work is needed to make these systems ready for real clinical use.

III. METHODOLOGY

A. DATA ACQUISITION AND DATASET CHARACTERISTICS

In this study, the CRIC Cervix Collection database [24] has been used, a rich archive of traditional cervical-vaginal cytology images acquired as a result of Pap smear analysis. The dataset contains actual clinical images which have been carefully categorized by expert cytopathologists as per the Bethesda System, the globally recognized standardized nomenclature of cervical cytology reporting. The main goal is to divide and categorize liquid-based cytology (LBC) images into three diagnostic categories which are clinically crucial: Negative for Intraepithelial Lesion or Malignancy (NILM), representing healthy cells with no threatening deformities, Low-Grade Squamous Intraepithelial Lesion (LSIL), indicating early mild cellular changes and High-Grade Squamous Intraepithelial Lesion (HSIL), which represents serious precancerous alterations that need an urgent medical intervention. The initial dataset had extreme imbalance between classes, where there were 6,779 NILM images, 1,360 LSIL images and 1,703 HSIL images equating to a total of 9,842 images. Such high imbalance and especially the underrepresentation of high-grade lesions present great difficulties to train the models because it may result in biased prediction along the majority of classifications, and poor generalization on minority classifications that are vital in clinical decision-making.

B. ADDRESSING CLASS IMBALANCE THROUGH CONDITIONAL GAN AUGMENTATION

In order to curb the extreme imbalance in the classes and to have a sound model training among all the diagnostic types, Conditional Generative Adversarial Network (CGAN) augmentation was adopted. CGANs, particularly compared to more traditional GANs, were selected because they can produce class-specific synthetic images given diagnostic labels such that the samples produced have the morphological properties that are unique to each lesion grade. The CGAN architecture entails the generator network G that is trained to recreate real cervical cells and a discriminator network D that is trained to differentiate real and synthetic images using the adversarial minimax objective function as shown in equation 1:

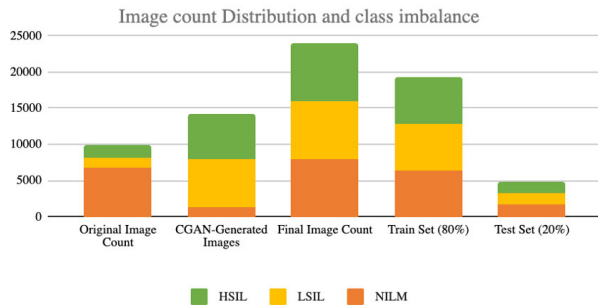
$$\min_G \max_D V(D, G) = \mathbb{E}_{x \sim p_{data}(x)} [\log D(x|y)] + \mathbb{E}_{z \sim p_z(z)} [\log(1 - D(G(z|y)))] \quad (1)$$

where x is represented as real images, and y as class labels (NILM, LSIL, HSIL), z is noise and $G(z|y)$ is the conditional, class-specific generation.

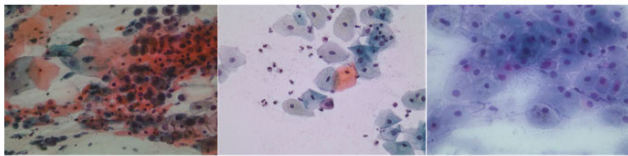
This adversarial training is carried out until the generator forms samples that cannot be distinguished with real clinical images. CGAN augmentation produced 1,221 more NILM, 6,640 more LSIL, and 6,297 more HSIL images, which

TABLE 1. Table showcasing the number of images per class before and after CGAN Augmentation.

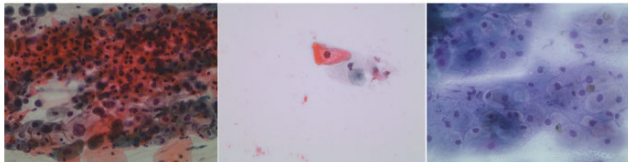
Class	Original Image Count	CGAN-Generated Images	Final Image Count	Train Set (80%)	Test Set (20%)
NILM	6,779	1,221	8,000	6,400	1,600
LSIL	1,360	6,640	8,000	6,400	1,600
HSIL	1,703	6,297	8,000	6,400	1,600
Total	9,842	14,158	24,000	19,200	4,800

**FIGURE 3.** Bar graph depicting the number of images per class before / after CGAN Augmentation and after train-test split.

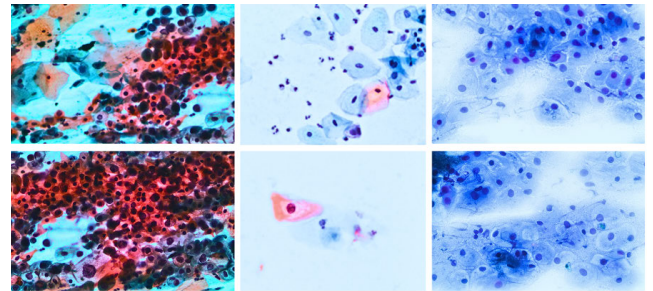
Sample images before C-GAN Augmentation:



C-GAN Augmented Images:

**FIGURE 4.** A comparison between original and synthetic images generated by CGAN.

brought each class to an equal number of 8,000 images as shown in Table 1. A graphical version of Table 1 is also represented as Fig. 3. Images produced by CGAN are truly faithful to the original cytological specimens and have realistic cell morphology, nuclear features, and staining patterns which are very similar to the true Pap smear images. A comparison between the original and the synthetic images has been shown in Fig. 4. An 80:20 stratified split was then used to split this balanced dataset (containing both real and CGAN-augmented images) into a training set (19,200 images) and a testing set (4,800 images). We acknowledge that evaluating an internal test set containing CGAN-generated samples carries a theoretical risk of data leakage. However, to definitively confirm that no synthetic leakage inflated our clinical metrics, we rely on the strict, real-only external validation presented in Section III-G. The model's sustained high performance on this completely independent, non-synthetic cohort (Brown Multicellular ThinPrep Database) guarantees that CERVIA learned true biological morphological features rather than memorizing synthetic data artifacts. Furthermore,

**FIGURE 5.** Images from the dataset after undergoing all the pre-processing methods.

the morphological consistency of these generated samples was implicitly validated through the model's robust performance on the independent external dataset. The high generalization accuracy confirms that the CGAN preserved critical diagnostic features—such as nuclear-cytoplasmic ratios and staining patterns—without introducing non-clinical artifacts that would have hindered performance on real-world data.

C. DATA PREPROCESSING AND AUGMENTATION

In addition to CGAN-based balancing between classes, the training set was augmented exclusively to increase the robustness of the model and its generalization skills. The augmentations add new controlled variations that depict real-world imaging conditions and inter-sample variations that are met in clinical practice. Brightness changes of $\pm 10\%$ were used to correct the differences in illumination of the microscopy, rotation changes of ± 45 degrees were used to address the arbitrary positions of cells in slide preparations, shift of 10 pixels was used in response to positional variability, and contrast changes of $\pm 30\%$ were used to simulate staining differences across labs. The Gaussian noise iterations were also used to simulate sensor artefacts, and optical blemishes. Sharpening was used to simulate focal plane variability in microscopic imaging. All these preprocessing methods are needed to make sure that the model learns aspects of the image that are invariant to technical differences and to avoid overfitting to particular imaging conditions in the training data. Fig. 5 shows some sample images after applying all the data pre-processing techniques. These images are used in model training.

D. MODEL ARCHITECTURE DESIGN

This paper presents CERVIA, a new hybrid deep-learning model which is a strategic blend of InceptionResNetV2 and MobileNetV3 models made by strategic fusion of features.

InceptionResNetV2 was chosen due to its outstanding performance of capturing multi-scale hierarchical features by utilizing inception modules to process multiple receptive field sizes of an input at the same time combined by residual connections that allow gradient flow during training. This architecture is very effective in detecting complex morphological features across various spatial scales that can range from fine nuclear structures to large scale cellular structures. The complementary architecture chosen to use is MobileNetV3 because it's an efficiency-optimized model utilizing depth wise separable convolutions and squeeze-and-excitation blocks to extract discriminative features in a fashion that is incredibly lightweight, yet capable of detecting subtle morphological distinctions for texture analysis. By incorporating MobileNetV3, the overall architecture maintains a balance between deep feature extraction and computational efficiency. This design ensures the model remains suitable for clinical deployment, achieving rapid inference times (approximately 0.15s per sample) as demonstrated in the clinical report generation module. CERVIA's architectural innovation is the combination of features, by concatenating the learned deep features of both networks to build an integrated representation as shown in eq. 2:

$$F_{\text{fused}} = \text{Concat}(F_{\text{InceptionResNetV2}}, F_{\text{MobileNetV3}}) \in \mathbb{R}^{d_1+d_2} \quad (2)$$

where $F_{\text{InceptionResNetV2}} \in \mathbb{R}^{d_1}$ and $F_{\text{MobileNetV3}} \in \mathbb{R}^{d_2}$ are the feature vectors pulled out of each network, and $F_{\text{fused}} \in \mathbb{R}^{(d_1+d_2)}$ is the fused feature representation benefiting from the multi-scale recognition of InceptionResNetV2 and the lightweight fine-grained discrimination of MobileNetV3. This dual, architecture scheme realized a theoretical complementarity: InceptionResNetV2 harnesses deep inception modules to bestow global, multi, scale structural hierarchies (which are the very components of the cellular boundaries and clusters), whereas MobileNetV3 utilizes highly efficient depth, wise separable convolutions to fetch local high, frequency, fine, grained texture details (which are basically the chromatin granularity). Instead of element, wise operations (e.g., addition), simple concatenation was chosen to preserve the entire feature manifold of both networks and to give the subsequent dense layers the freedom of learning the optimal weighting between these structural and textural features automatically without any loss of information.

E. TRAINING CONFIGURATION AND OPTIMIZATION

The pre-training of the integrated CERVIA model adopted the Adam optimizer, owing to its adaptive learning rate and its ability to handle sparse gradients effectively—both features particularly useful in medical imaging applications. To ensure steady convergence, a conservative learning rate of 10^{-7} was adopted, taking into account the deep structure of the model and the sensitivity of medical classification tasks, where incremental weight updates can induce forgetting yet facilitate the learning of fine-grained features. Training was

conducted for a total of 90 epochs to ensure the model learned distinguishing features for every class, while the model's performance on the validation dataset was constantly monitored to preclude overfitting. Considering the nature of the three-class challenge, a binary cross-entropy loss function was adopted by redefining the task as multiple binary classifications, as shown in Eq. (3):

$$L_{BCE} = -\frac{1}{n} \sum_{i=1}^n [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (3)$$

Here, n is the number of samples, y_i the true label, and $\hat{y}_i$ the predicted probability of the i -th sample. It effectively handles the probabilistic nature of the predictive outputs in the diagnosis task and provides smooth gradients for optimization.

F. FINE-TUNING STRATEGY

After primary training, fine-tuning was conducted to enhance the model's discriminative ability for cervical lesion classification. Initially, all layers were frozen to retain low-level feature representations, and only the final 50 layers were made trainable. This approach enabled the model to update high-level decision boundaries specific to cervical cytology while ensuring that strong generic feature extractors learned during the primary training were retained. The learning rate was lowered to 10^{-9} to allow subtle adjustments of the weights during fine-tuning without disturbing the learned representations. Training proceeded for 50 additional epochs using the same binary cross-entropy loss function, while the Adam optimizer maintained adaptive learning rates for each trainable parameter. This tuning stage produced the final CERVIA model, which achieved 97.1% accuracy on the CRIC test set. A model architecture diagram illustrating all steps is presented in Fig. 6.

To ensure strict replicability and deterministic results across experiments, the following parameters and configurations were maintained:

- **Random Seed:** A fixed global seed of 42 was utilized across all computational libraries for data splitting, shuffling, and weight initialization.
- **Preprocessing Pipeline:** As detailed in Section III-C, images underwent targeted augmentation ($\pm 10\%$ brightness, $\pm 45^\circ$ rotation, 10-pixel positional shifts, $\pm 30\%$ contrast, Gaussian noise injection, and focal sharpening) applied exclusively to the training set to prevent test leakage.
- **Data Split:** 80% Training / 20% Testing (stratified by class).
- **Optimizer & Loss:** Adam optimizer ($\beta_1 = 0.9$, $\beta_2 = 0.999$) paired with a Binary Cross-Entropy loss function.
- **Learning Rates & Epochs:** Primary training utilized a learning rate of 10^{-7} for 90 epochs, followed by a fine-tuning phase at 10^{-9} for 50 additional epochs.
- **External Validation:** Zero-shot inference conducted on 200 independent images from the Brown ThinPrep

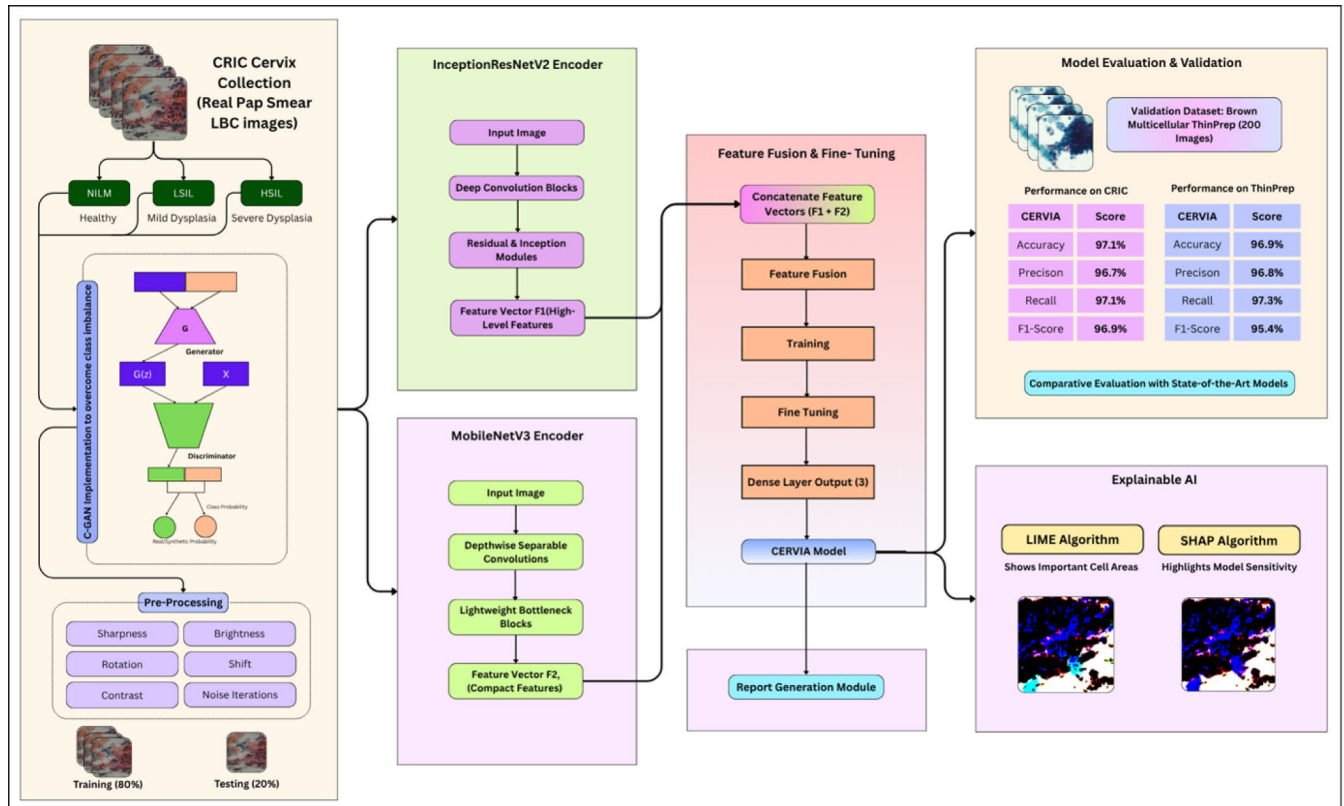


FIGURE 6. CERVIA model architecture diagram.

Database, with no prior re-training or domain adaptation.

G. EXTERNAL VALIDATION AND CROSS-DATASET GENERALIZATION

An independent validation of the model was also performed to estimate its generalizability outside the training distribution through the Brown Multicellular ThinPrep Database [25], consisting of 200 independent LBC images, which are drawn from another clinical source and potentially utilize different staining protocols, imaging systems, and patient populations. CERVIA framework produced 96.9% accuracy on this independent data set, showing substantial strength on various data sets and suggesting the ability of the features learned by CERVIA to be extrapolated successfully to various clinical populations and imaging scenarios, which represents an essential requirement of real-world testing.

H. EXPLAINABLE AI INTEGRATION

Two explainable AI (XAI) methodologies were incorporated to foster clinical transparency and allow the pathologist to confirm automation outputs: Local Interpretable Model-agnostic Explanations (LIME) and SHapley Additive exPlanations (SHAP). LIME was chosen because it produces understandable local predictions by highlighting parts of the image most influential for any given prediction, and SHAP employs the theory of cooperative game to estimate the

importance of features globally and assigns contributions to each pixel for any given prediction. Such XAI strategies are crucial to clinical practice, as they give pathologists the ability to confirm model reasoning, reveal errors, and build confidence in automation-based diagnostic systems.

I. PERFORMANCE EVALUATION AND CLINICAL REPORTING

Confusion matrices, accuracy and ROC curves were utilized to examine model performance thoroughly. It was aligned to state-of-the-art approaches and relative comparison was performed to identify CERVIA's competitive advantages. To convert model's outputs into diagnostic action, the clinical report generation module was developed which integrates the original LBC image, classification result along with XAI overlay figures, cell-wise distribution, qualitative morphological analysis and final diagnostic interpretation, providing the entire transparent pipeline to the clinicians.

IV. RESULTS AND ANALYSIS

The comprehensive assessment of CERVIA demonstrates outstanding performance in automated cervical cytology classification by presenting superior accuracy across all diagnostic categories when compared to standalone architectures and existing state-of-the-art methodologies. The structured evaluation comprises training dynamics analysis, classification performance evaluation, comparative benchmarking,

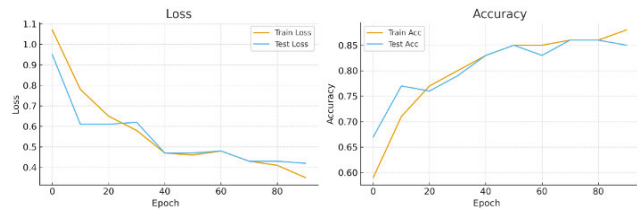


FIGURE 7. Training and convergence curves of CERVIA during initial training.

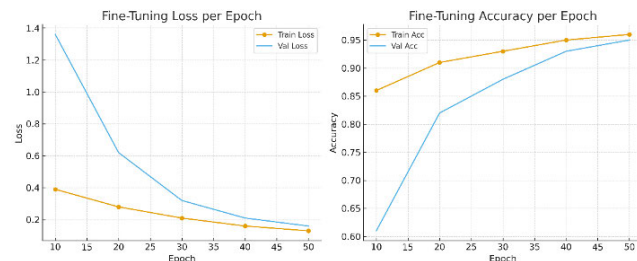


FIGURE 8. Fine-tuning performance showing improved accuracy and stability.

ablation studies, and explainable AI validation that have all been combined to establish the clinical viability and diagnostic reliability of the proposed hybrid framework.

A. MODEL TRAINING AND CONVERGENCE ANALYSIS

The optimization of the CERVIA model through its initial and fine-tuning training phases was systematically documented, revealing stable optimization and consistent convergence behavior. During the first training phase, which lasted for 90 epochs, the model exhibited regular and smooth optimization, as illustrated in Fig. 7. The training loss gradually decreased from approximately 1.05 to 0.1, while the test loss reduced from about 1.0 to 0.4. Both losses followed a similar trend and remained closely aligned throughout most of the training process. This close correspondence indicates strong generalization capability and the absence of overfitting, demonstrating that the model learned robust features rather than memorizing training-specific patterns. Simultaneously, the training accuracy increased steadily from around 60% to 87%, while the test accuracy followed a similar trajectory, reaching a peak of 85–86%. These parallel trends in accuracy curves confirm stable learning dynamics and validate the appropriateness of the chosen hyperparameters, including the conservative learning rate of 10^{-7} combined with the Adam optimizer. This stage effectively finalized the training of the feature extractors in the InceptionResNetV2 and MobileNetV3 branches, forming a strong foundation for subsequent fine-tuning and preventing potential performance degradation from early saturation or divergence.

The fine-tuning phase resulted in rapid convergence and high classification accuracy, as illustrated in Fig. 8. After unfreezing the last 50 layers and reducing the learning rate to 10^{-9} , the training loss decreased stepwise from 0.39 to 0.08 over 50 epochs, while the validation loss reduced significantly from 1.35 to 0.15. This indicates

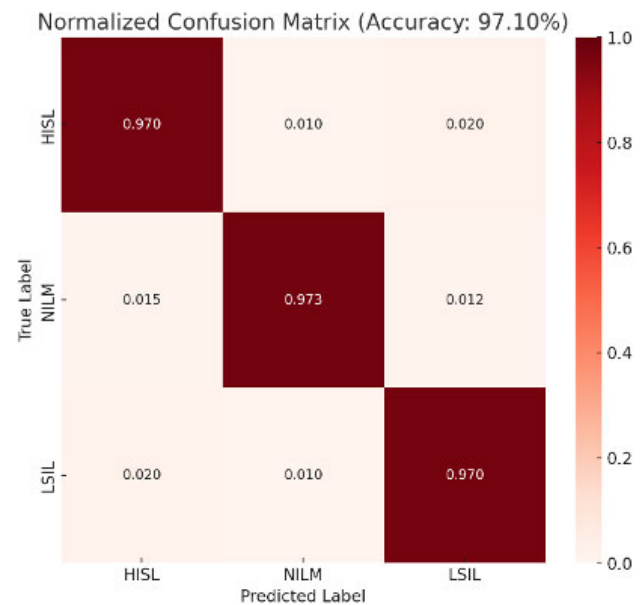


FIGURE 9. Normalized confusion matrix illustrating high class-wise accuracy.

an improved adaptation to dataset-specific morphological features essential for the accurate identification of NILM, LSIL, and HSIL categories. The training accuracy increased notably from 85% to 97.1%, while the validation accuracy rose sharply from approximately 65% to 94.8%. The near convergence of both loss and accuracy curves after 40 epochs indicates minimal overfitting and strong generalization capability, confirming that the fine-tuning strategy refined the decision boundaries without disrupting the previously learned representations. This stage highlights the effective use of layer unfreezing and differential learning rates in transfer learning, enabling the model to capture intricate cytological features that are crucial for precise diagnostic classification. The CERVIA model demonstrated superior diagnostic performance across all normal and abnormal categories, while maintaining training stability throughout the fine-tuning epochs.

B. CLASSIFICATION PERFORMANCE AND DISCRIMINATIVE CAPABILITY

The CERVIA model was impressive in its performance as it managed to accurately classify cases in the test data about 97.1% of the time, and its results across the three diagnostic categories were very consistent. The normalized confusion matrix in Fig. 9 expresses the dominance of the diagonal with the accuracy corresponding to each class of 97.0% for HSIL, 97.3% for NILM, and 97.0% for LSIL. The misclassifications were limited to only 1–2% across all classes, and no significant class bias is observed despite the presence of severe imbalance in the original dataset. This suggests that the CGAN-based augmentation strategy balanced the classes without including synthetic artifacts that might bias model predictions. This model confidently identifies high-grade

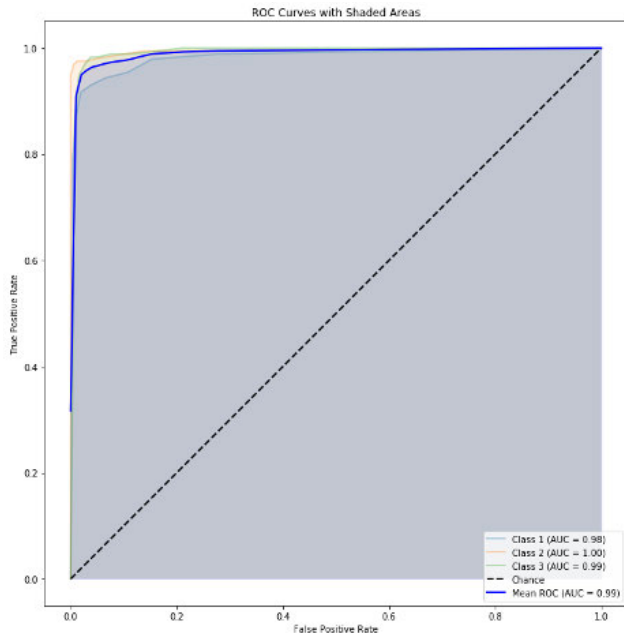


FIGURE 10. ROC curves showing excellent class separability and high AUCs.

and low-grade lesions and normal cytological samples with high sensitivity and specificity without compromising either recall or precision. Therefore, it is highly reliable in clinical diagnostics.

Despite the high overall accuracy, an analysis of the minor misclassifications (the 1–2% off-diagonal elements in the confusion matrix) reveals specific, predictable failure modes. Errors predominantly occurred in complex imaging scenarios characterized by heavily overlapping cell clusters or poor staining, which obscure defined nuclear boundaries. Furthermore, misclassifications between LSIL and HSIL often involved borderline or transitional lesions, where the nuclear-cytoplasmic ratio and chromatin granularity mimic characteristics of both stages. These failure modes highlight the inherent ambiguities in cervical cytology rather than systemic architectural flaws, further emphasizing the need for this system to act as an AI-assisted support tool for human cytopathologists rather than a fully autonomous replacement.

Further validation of the discriminative capability of CERVIA was done using the receiver operating characteristic analysis, which points to the superior class separability exhibited by the model. Fig. 10 presents the AUCs for each class: HSIL attained 0.98, NILM reached 1.00, and LSIL attained 0.99. The mean ROC curve shows an aggregate AUC of 0.99, located closer to the top-left ideal corner and far from the random baseline diagonal. This represents very good classification confidence with minimal overlap between diagnostic categories. High AUC values point to that the model has high true positive rates or sensitivity and true negative rates or specificity, thus CERVIA is very effective in limiting the number of false positives and false negatives in all classes. The ROCs are very smooth and this

is probably due to the classifier having very well-defined decision boundaries and being very stable in probability estimation for the different classes. This is indicative of the network's ability to produce reliable probability scores and stable predictions, proving that the hybrid feature fusion approach effectively enhances class discrimination beyond the results from individual architectures.

TABLE 2. Per-class performance metrics of the CERVIA model.

Metric	HISL	NILM	LSIL	Macro Avg
Precision	0.96	0.97	0.97	0.9667
Recall	0.97	0.973	0.97	0.9710
F1 Score	0.965	0.971	0.970	0.9687

More detailed per-class performance characteristics are shown in the classification report presented in Table 2. This summarizes the performance of the CERVIA along target classes HSIL, NILM, and LSIL with precision, recall, and F1-score metrics, with consistently high values, showing that it performs balanced and reliable classification. In particular, precision scores are as follows: 0.96 for HSIL, 0.97 for NILM, and 0.97 for LSIL, indicating very good accuracy in positive predictions, with a low number of false positives and proving that when the model predicts a particular class, it is highly likely to be accurate. Recall values of 0.97 for HSIL, 0.973 for NILM, and 0.97 for LSIL give evidence of the strong sensitivity of the model to effectively catch the true positives-to actually detect real cases of each of the lesion grades. Corresponding F1-scores amount to 0.965, 0.971, and 0.970, respectively, reflecting optimal balancing between precision and recall for each class and hence meaning the model keeps high performance without any single metric being favoured over another. Finally, the macro average F1-score of 0.9687 further confirms that the performance is uniform across all the diagnostic categories, indicating that CERVIA generalizes well without bias towards any class. These comprehensive metrics indicate the robustness, consistency, and reliability of the model for accurate real-world cervical cytology image classification in clinical settings.

C. COMPARATIVE ANALYSIS AND MODEL SUPERIORITY

The competitive advantages of CERVIA were established through comprehensive comparative analysis against contemporary state-of-the-art architectures and prior research efforts. Quantitative comparisons with benchmark deep learning models that were trained with the same datasets are presented in Table 3 and Fig. 11. The proposed hybrid architecture, InceptionResNetV2 + MobileNetV3, has achieved 97.1% for accuracy, 96.7% for precision, 97.1% for recall, and 96.9% for F1-score. This has significantly outperformed all competing architectures. MobileNetV3 alone has recorded 96.2% accuracy, trailed by Xception with 95.3%, InceptionV3 with 94.6%, DenseNet201 with 93.5%, and ResNet50 with 92.4%. An improvement of 1–1.5% along all the evaluation metrics substantiates the effectiveness

TABLE 3. Comparative evaluation of CERVIA with baseline architectures.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
<i>ResNet50</i>	92.4	91.8	92.2	91.9
<i>DenseNet201</i>	93.5	93.1	93.4	93.2
<i>InceptionV3</i>	94.6	94.2	94.5	94.3
<i>Xception</i>	95.3	95.0	95.1	95.1
<i>MobileNetV3</i>	96.2	95.8	96.1	96.0
<i>Proposed (InceptionResNetV2 + MobileNetV3)</i>	97.1	96.7	97.1	96.9

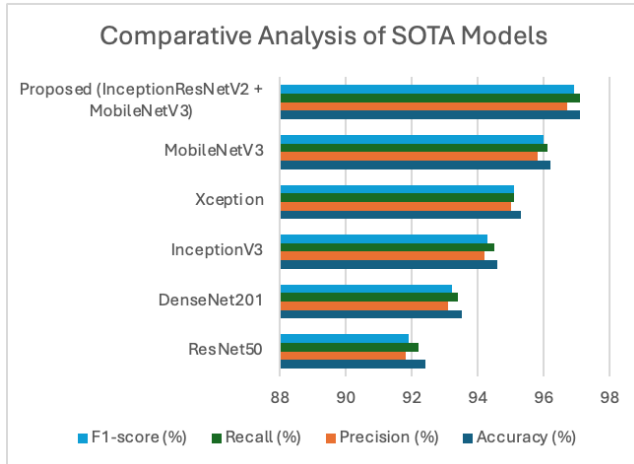


FIGURE 11. Comparison of CERVIA with baseline architectures.

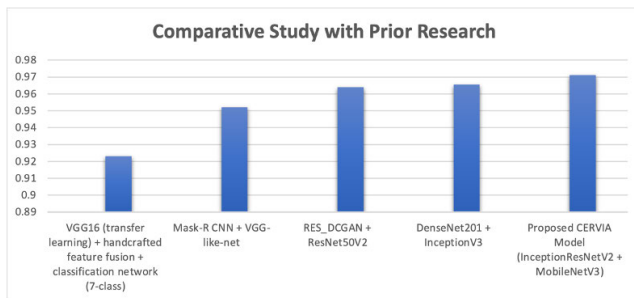


FIGURE 12. Performance comparison of CERVIA with prior research methods.

of the hybrid fusion approach in capturing complementary morphological features. Thus, the best trade-off between model complexity and classification accuracy corresponds to the proposed architecture, where a combination of hierarchical features from diverse backbone networks produces more informative and robust representations in fine-grained cervical cytology classification tasks. To ensure a rigorously fair and standardized comparison, all baseline architectures were trained from scratch on the exact same augmented dataset and evaluated under identical experimental conditions, hyperparameter configurations, and evaluation metrics as the proposed CERVIA model.

While exhaustive hardware-specific inference benchmarking across all baseline architectures falls outside the current scope, the deployment efficiency of CERVIA is structurally

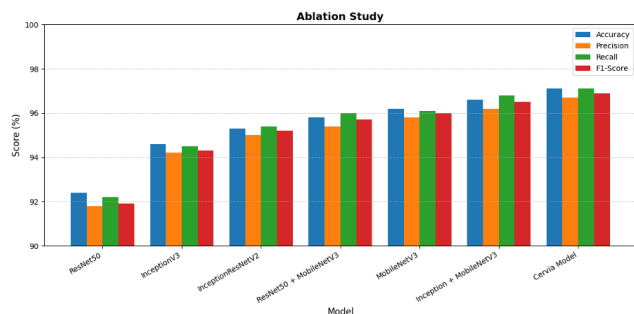
evident. By integrating the lightweight MobileNetV3—which utilizes parameter-efficient depth-wise separable convolutions—the hybrid model avoids the massive computational overhead and latency bottlenecks typical of heavier monolithic baselines like DenseNet201. In practical deployment, as demonstrated by our clinical reporting module, CERVIA achieves a rapid inference time of approximately 0.15 seconds per sample. This confirms that the model not only sets a new benchmark in accuracy but also maintains the high-throughput efficiency required for real-time clinical screening environments.

Comparison with prior research efforts further cements the superiority of CERVIA in this domain. Fig. 12 shows that CERVIA outperforms most prior research work, such as the VGG16-based model proposed by Hao et al. [26], which achieved an accuracy of 92.3%. The Mask R-CNN + VGG-like network proposed by Kurnianingsih et al. [27] attained 95% accuracy, the RES_DCGAN + ResNet50V2 model proposed by Wubineh et al. [4] achieved 96.4%, and the DenseNet201 + InceptionV3 hybrid framework by Sharma et al. [10] achieved 96.54%. These results demonstrate an improvement of approximately 1–5% over strong prior models. While earlier methods primarily relied on hand-crafted features or computationally heavy architectures, the proposed hybrid framework strikes a balance between deep contextual learning and computational efficiency through its dual-branch design, which captures multi-scale semantic features along with efficient low-latency representations. Such a considerable improvement in performance validates the superiority of the fusion approach and establishes a new benchmark for automated cervical cytology classification with enhanced accuracy, superior generalization capability, and improved deployment efficiency compared to current state-of-the-art methodologies.

Extensive ablation studies were performed, which systematically validate the contribution of each architectural component by looking at various model variants and combinations. Table 4 and Fig. 13 show ablation analysis through the impact of different backbone configurations on the accuracy, precision, recall, and F1-score of models. Among all the configurations tested, the CERVIA model, representing InceptionResNetV2 + MobileNetV3, achieved the top scores in all metrics. Individual backbone architectures such as ResNet50 (92.4%), InceptionV3 (94.6%), and InceptionResNetV2 (95.3%) had strong but relatively lower performances that emphasize the idea that standalone

TABLE 4. Ablation study showing performance across model variants.

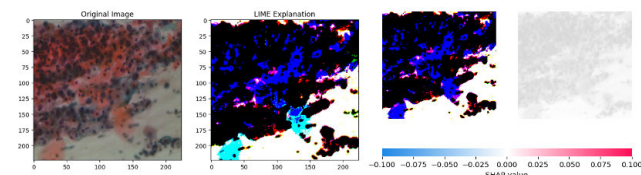
Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
<i>InceptionResNetV2</i>	95.3	95.0	95.4	95.2
<i>Inception + MobileNetV3</i>	96.6	96.2	96.8	96.5
<i>ResNet50 + MobileNetV3</i>	95.8	95.4	96.0	95.7
<i>InceptionV3</i>	94.6	94.2	94.5	94.3
<i>ResNet50</i>	92.4	91.8	92.2	91.9
<i>MobileNetV3</i>	96.2	95.8	96.1	96.0
<i>CERVIA Model</i>	97.1	96.7	97.1	96.9

**FIGURE 13.** Ablation study showing the effect of different backbone combinations.

architectures capture only partial feature hierarchies and do not have the comprehensive representational capacity achieved through their fusion. The intermediate combination of residual networks, such as ResNet50 + MobileNetV3 (95.8%) and Inception + MobileNetV3 (96.6%), shows obvious benefits due to the combination of complementary networks however they still underperform compared to the proposed fusion of InceptionResNetV2 + MobileNetV3. The progressive improvement trend across variants strengthens the notion that feature fusion considerably enhances the representational power and generalization capability. In the meantime, it can be seen that using MobileNetV3 alone produced an accuracy of 96.2%, which proves its efficiency as a strong, standalone architecture. At the end, the ablation study confirms that integrating deep residual inception modules of InceptionResNetV2 with efficient depthwise separable convolutions of MobileNetV3 produces the most balanced and high-performing architecture among all the tested variants, validating the architectural design choices underlying CERVIA.

D. EXPLAINABLE AI ANALYSIS AND MODEL INTERPRETABILITY

Comprehensive explainability analysis with both LIME and SHAP was done on a wide range of cytological samples to provide clinical transparency that would allow the pathologist to verify automated predictions. These two complementary XAI methods play a vital role in unlocking the black box of deep learning models, which is a major problem that hinders clinical adoption. The authors systematically compared the two methods for their consistency, biological relevance,

**FIGURE 14.** LIME vs SHAP visualizations for Sample 1 highlighting nuclear focus.

and diagnostic utility on six representative samples that covered the range of cellular density, nuclear arrangement, and lesion grades.

The first sample shown in Fig. 14 consisted of densely packed epithelial cells with small nuclei and cytoplasmic texturing, which served as a good example for the two explainability approaches to highlight their differences. The work of LIME in the explanation demonstrates that the cyan and magenta regions correspond to influential superpixels however these activations seem to be somewhat scattered over the background and non-relevant areas. This means that although LIME identifies larger contextual regions, it does not localize cellular nuclei and borders around which are the main diagnostic features in cytopathology. In contrast, the SHAP heatmap contains more organization and interpretation of activations, with the dark blue areas corresponding to the most negatively contributing zones, and the bright red patches representing the strongest positively contributing features. These nuclear clusters and cytoplasmic borders are very well aligned. The grayscale SHAP representation that goes along with it shows that the focus is indeed on the densely packed cellular regions with the retention of the morphological detail which is very important for the grading of the lesion. Thus, the pixel-level explanation which is provided here by SHAP brings out biologically relevant features like nuclear density, shape irregularities, and chromatin texture, all of which are significant factors in cytological diagnosis.

The second sample shows the same trends, with a large number of epithelial cells that are partially overlapping and have dense nuclear clustering (Fig. 15). The active areas of LIME, in contrast, are somewhat disjointed in character with their cyan and red colors and they also spread into the background thus indicating that too many segments have been made. While LIME points out large cellular contexts, it does not distinctly mark the diagnostically relevant nuclei leaving the surrounding non-informative areas still there

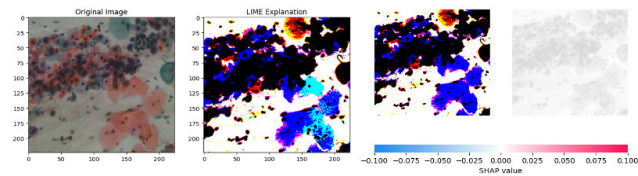


FIGURE 15. LIME vs SHAP visualizations for Sample 2 showing dense nuclear clustering.

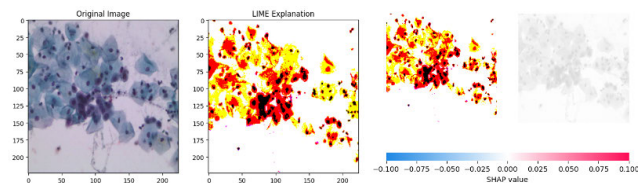


FIGURE 16. LIME vs SHAP visualizations for Sample 3 emphasizing nuclear boundaries.

and thus the interpretive focus is lowered. In contrast, the SHAP heatmaps show more exact positions of the red regions indicating the nuclei that have the most influence and the blue regions showing the negative contributions of the cytoplasmic background. Such emphasis on compact nuclear groups is evidence of the presence of discriminative features that SHAP offers. The grayscale SHAP also supports the dense activations being aligned with the morphological detail.

The third sample comprising of squamous epithelial cells with clearly defined nuclear boundaries and the cytoplasm being lightly stained further brings out the weakness of LIME (Fig. 16). LIME highlights several superpixel segments in yellow and red, distributed across cytoplasmic regions, showing over-segmentation with activation of broad, diffuse areas that include non-critical background pixels. Isolating specific morphological cues (nuclear shape, chromatin density, boundary texture) that drive classification decisions is hard. SHAP provides a tighter, biologically aligned focus whereby high positive SHAP regions are concentrated around the nuclear centers and cell borders. Negative zones correspond to cytoplasmic or unstained areas. This clearly shows that the decisions taken by CERVIA were indeed strongly dependent on nuclear compactness and chromatin texture, while providing accurate, interpretable feature localization consistent with cytological diagnostic protocols.

Samples four and five (Figs 17 and 18) demonstrate consistent superiority of SHAP across varying cellular arrangements. However, in the case of a densely packed epithelial cell image, LIME present significant overlap with irrelevant background regions and fragmented segmentation that decreases interpretive clarity (Fig. 11). On the other hand, SHAP presents targeted activations where bright red zones correspond exactly to densely stained nuclear areas that allow fine-grained localization aligned with diagnostic morphological cues. The sample of well-separated nuclei, for which LIME presents extensive red and yellow regions extending into irrelevant background areas to capture contextual color gradients rather than strictly morphological cues, is shown in

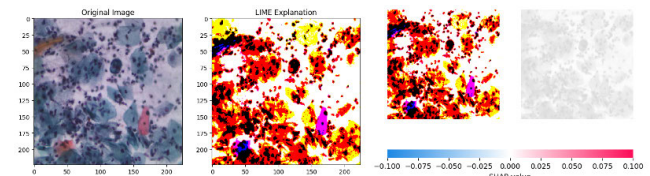


FIGURE 17. LIME vs SHAP visualizations for Sample 4 with densely packed nuclei.

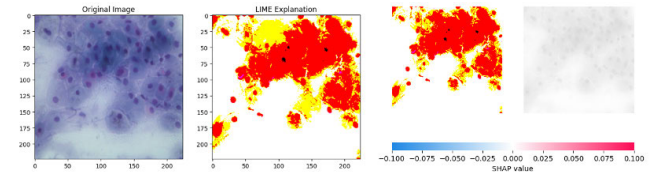


FIGURE 18. LIME vs SHAP visualizations for Sample 5 showing isolated nuclei.

Fig. 12. Compared to that, the activation shown in the SHAP image is very close to each nucleus and points to the places that had the greatest impact on the model's prediction, thus, the outlines of the nuclei as well as the denser parts of the chromatin being traced most clearly.

The last sample displaying regular rows of epithelial cells with sharp nuclear contours is another instance of this tendency being confirmed (Fig. 19). LIME produces patchy and overlapping activations that go further into less relevant background regions. These highlight generalized differences in intensity, rather than specific nuclear shapes. SHAP offers selective and morphologically relevant explanation. Notice the red regions align precisely with individual nuclei and cellular boundaries. This suggests that the model classification relies primarily on nuclear morphology and texture consistency which are hallmarks of diagnostic importance.

Over all samples analysed, SHAP consistently provides more clear and biologically relevant interpretability than does LIME. Moreover, SHAP pixel attribution aligns well with expert diagnostic visual cues on nuclear prominence, chromatin texture, membrane definition, and cellular density-the very morphological features a cytopathologist considers during manual screening. LIME is useful to identify broader contextual regions but suffers from over-segmentation, inclusion of background noise, and limited precision in isolating cellular structures critical for diagnosis. Theoretically, such a top-level performance of SHAP over LIME is based on the different mechanisms of feature attribution in relation to the cytopathological criteria.

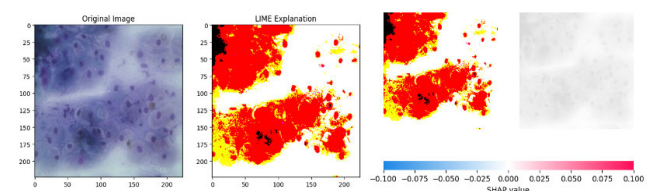


FIGURE 19. LIME vs SHAP visualizations for Sample 6 confirming SHAP's precision.

The diagnosis of dysplasia (LSIL/HSIL) is largely a differential one made on the basis of the Nuclear-Cytoplasmic (N:C) Ratio and nuclear chromatin texture.

As cooperatives game theory based approach (Shapley values) SHAP calculates the marginal contribution that belongs to a single pixel. Thus the model is able to attribute very high positive importance to the nuclei and negative or neutral importance to the cytoplasm only, thus it very precisely and effectively corresponds to the N:C ratio.

LIME, on the other hand, uses superpixel segmentation to create local perturbations. If the boundary of a superpixel is not perfectly aligned with the biological boundary of the cell which is most likely in case of complicated overlapping smear then the importance is “spread out” between the relevant nuclei and the irrelevant background.

This theoretical shortcoming of LIME accounts for the “over-segmentation” and background interference seen in the LIME (Figs. 14–19) and SHAP’s pixel-wise independence is naturally consistent with the fine morphology required for grading nuclear aberrations.

Thus, SHAP is the superior method of explainability for CERVIA since it will give reliable and clinically meaningful feature attributions necessary for pathologist trust and model validation during clinical deployment.

E. REPORT GENERATION AND INTERPRETABILITY MODULE

The CERVIA reporting module automatically generates structured, interpretable, and clinically oriented reports from AI-assisted cytological analysis. Each report includes the usual diagnostic details which include things like patient info, slide and report IDs, the version of the model used, when the prediction was made, and the threshold applied during inference. This setup makes it easier to track results and verify them later for clinical checks or regulatory review. A visual of the sample report generated by CERVIA model can be seen in Fig. 20.

At the heart of the report is a short summary generated with help from the AI system. It shows the final diagnosis for example, NILM, LSIL, or HSIL and also gives a confidence score that shows how sure the model was about its decision. For example, a “NILM-80% confidence” would reflect probabilistic belief by the system in that classification, not to be confused with a clinical probability estimate. The report highlights that the AI-generated results should be confirmed by human cytopathologists for the diagnostics to be reliable.

The most significant feature of the network is the usage of XAI visualizations, which improve user understanding and openness. The system comprises two types of visual overlays. The initial one is a SHAP heatmap, which indicates the exact pixel areas that have the most significant influence on the model’s decision. The second is a LIME heatmap, which clarifies the wider context by illustrating how larger areas, or superpixels, contribute to the prediction. Thus, the expert user can, with the help of visuals, check whether the AI model is using the right morphological factors such as nuclear

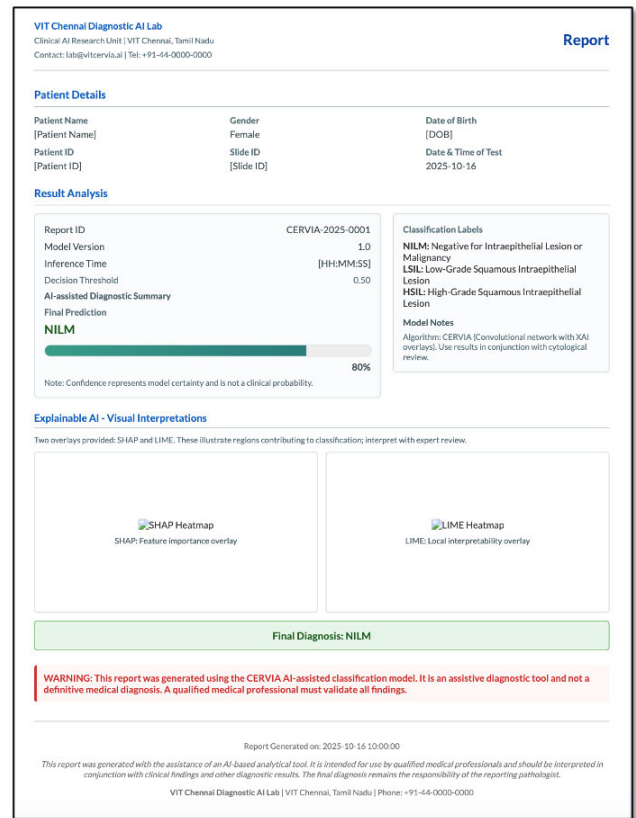


FIGURE 20. CERVIA report generation interface with SHAP and LIME overlays.

boundary, chromatin texture, and cytoplasmic patterns for making the decision.

This component takes the results from the module and automatically converts them into a uniform format suitable for printing in PDF or HTML form that can be read by hospital and laboratory information systems. Consequently, the CERVIA report creation tool, which provides a transparent and clinically feasible connection from deep learning inference to expert cytological confirmation, is basically a single user interface where quantitative predictions, XAI visual evidence, and medical disclaimers are embedded.

V. CONCLUSION

The article describes CERVIA, a novel hybrid deep learning architecture for the automatic classification of cervical cancer that, by design, resolves problems of computational cytopathology raised by severely imbalanced classes, limited interpretability, and diagnostic categories aligned with clinical practice. The principal contribution of this work is the deliberate combination of the InceptionResNetV2 and MobileNetV3 architectures, which in a complementary manner unite the hierarchical multiscale feature extraction with the quite-resolution pattern recognition abilities. The Conditional GAN-powered data augmentation takes care of the problem of extreme underrepresentation of high-grade lesions by producing 14,158 synthetic images so that all

classes can be balanced to 8,000 samples each and at the same time the model is trained robustly and unbiased. Moreover, the use of two interacting explainable AI methods, LIME and SHAP, provides essential transparency by focusing on the morphological features most relevant for the diagnosis; SHAP ensures the best pixel-level interpretability most closely matching the expert assessment criteria in cytopathology. The all-inclusive clinical report generation module converts model predictions into practical diagnostic guidance, thus constituting an end-to-end transparent framework ready for clinical translation.

Measured in numbers, CERVIA got a 97.1% accuracy, 96.7% precision, 97.1% recall, and 96.9% F1-score on the CRIC test set and was better than the single architectures such as ResNet50 (92.4%), InceptionV3 (94.6%), and MobileNetV3 (96.2%). This model was nearly perfect in terms of ROC-AUC, as it achieved 0.98 for HSIL, 1.00 for NILM, and 0.99 for LSIL, with the average AUC being 0.99. The independent Brown Multicellular ThinPrep Database was used for the external validation that resulted in 96.8% accuracy, thus demonstrating strong cross-dataset generalization. These performance gains consequently set the state-of-the-art benchmark at 1-5% higher than those of the previously reported methods.

CERVIA modernizes in a fundamental way the workflow of cervical cancer screening which used to be a labor intensive manual process and is now converted into an efficient automated system. As per the conventional method, cytopathologists are required to manually check the Pap smear slides under a microscope, visually detecting abnormal cells among thousands of normal cells this activity consumes 10-15 minutes per slide, demands a lot of training, and is vulnerable to mistakes caused by fatigue and differences in the opinion of various observers. On the other hand, CERVIA is capable of scanning a high resolution cytology image in just a few seconds, it also automates the identification and classification of cells into NILM, LSIL, and HSIL categories at 97.1% accuracy, and at the same time, provides the visualizations that can be understood by the user which feature the most diagnostically relevant aspects. The machine prints detailed diagnostic reports that pool together the classification outcomes, confidence levels, XAI overlays, and the analysis of the distribution of cells on a per cell basis which helps the pathologists to quickly check the predictions and also decide where to put their attention by identifying the most potential cases. It not only cuts down the screening time by close to 80%, but also lessens the subjective interpretation bias, guarantees the maintenance of consistent diagnostic standards in different facilities, and makes it possible to deploy them in areas with limited resources where there is a lack of specialized expertise finally, it results in faster diagnosis, lower costs, and better patient outcomes due to the fact that the disease is caught earlier and therefore intervention can be done at an earlier stage.

Besides these successes, it is worth mentioning some limitations. Firstly, although CGAN augmentation effectively

resolves the issue of class imbalance, synthetic images may not perfectly represent the entire range of morphological variability due to diverse populations, thus potentially causing minor artifacts. Secondly, the hybrid architecture, although it is a robust one, requires a moderate level of computational power which should be optimized for edge deployment in resource, limited environments.

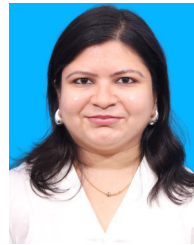
Moreover, the present framework is based only on cytological image data. Future versions might gain from multi-modal integration, for example, by including patient clinical history and HPV status, to further raise the level of diagnostic accuracy.

Future studies should be directed at the extension of validation to larger, multi-institutional datasets representing diverse demographic populations and different imaging conditions in order to strengthen generalizability. The investigation of lightweight model compression techniques such as knowledge distillation and pruning could make deployments on edge devices for point-of-care screening in resource-limited settings a reality. Extending the classification framework to include other Bethesda System categories such as ASC-US and AGC would provide richer diagnostic coverage. By integrating multi-modal data such as a patient's clinical history and results of HPV testing, the diagnostic accuracy may be enhanced through the fusion of complementary information. Finally, prospective clinical trials with real-time validation by pathologists will confirm the utility of the clinic and implementation barriers essential for successful translation into routine clinical practice. CERVIA is a significant step in the direction of screening systems that are accessible, reliable, and explainable, for cervical cancer, and it has the potential to decrease diagnostic accuracy disparities and eventually save lives in the different healthcare environments worldwide. Additionally, future work will focus on a rigorous clinical validation framework, inviting cytologists to perform blind evaluations on synthesized images to quantify their similarity to real samples in terms of specific morphological metrics.

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